

Figure 3: The folder structure of the Ionic app



Figure 4: The Ionic app development process

These steps are as illustrated in Figure 4.

Ionic app testing methods

The app built using Ionic can be tested using the variations shown in Figure 5.

Desktop browser: This method allows you to test your app in the desktop browser using the following command, in the app directory:

```
$ ionic serve
```

This command would trigger the live-reload server for the project, which means that any changes made to the underlying HTML, CSS or JavaScript will be directly reflected in the result.

Simulator test: The corresponding platform's simulator can be utilised to test the app built using the Ionic framework. The *build*, *emulate* and *cordova prepare* commands are employed.

Mobile browser: The app can be directly tested in a mobile browser. The desktop test server running the application can be accessed from a mobile Web browser to test the application. This type of app testing is only suitable for simple apps. When the apps get fairly complex, the behaviour is interrupted by some of the browser features.

Native app testing (device): Another method to test the app is as the native app in the device. The device needs to be plugged in, and the following command should be executed to directly test on the device:

```
$ ionic run android
```

A simple app

The JavaScript portions of the Ionic app are built using *Angular.js*. A List app with portions of code in *index.html*, *app.js* and *controllers.js* are given below:

```
<scriptsrc="js/app.js"></script>
```

```

<scriptsrc="js/controllers.js"></script>
</head>
<body ng-app="starter">
  <ion-pane>
    <ion-header-bar class="bar-royal">
      <h1 class="title">OSFY Ionic Demo</h1>
      <div class="buttons">
        <button class="button" ng-click="openModal()">
          Add
        </button>
      </div>
    </ion-header-bar>
    <ion-content>
      <ion-list ng-controller="ToDoListCtrl">
        <ion-item ng-repeat="item in toDoListItems">
          {{item.task}}
        </ion-item>
      </ion-list>
    </ion-content>
  </ion-pane>
</body>
In app.js, type:
angular.module('starter', ['ionic', 'starter.controllers'])
In controllers.js, type:
angular.module('starter.controllers', [])
  .controller('ToDoListCtrl', function ($scope) {
$scope.toDoListItems = [
{
  task: 'Open Source For You',
  status: 'not done'
},
{
  task: 'Electronics For You',
  status: 'not done'
}]
});

```



Figure 5: Ionic app testing methods

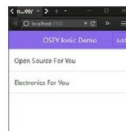


Figure 6: Ionic app demo

The output of the above code is shown in Figure 6.

The Ionic Lab

The Ionic Lab enables developers to harness the power of Ionic with the help of a desktop app. It allows developers to